Credit Scorecard Development and Validation Report

Retail Personal Loan Scorecard · Unsecured personal loans (synthetic example)

Prepared for	Demo Client
Engagement	DEMO-0004
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Executive summary

Demonstration report. Built entirely from synthetic data generated by ProQuant for illustration. No real lender, applicants or loans are used.

ProQuant Analytics developed and independently validated the **Retail Personal Loan Scorecard** for Demo Client (Unsecured personal loans (synthetic example)): a logistic-regression scorecard with a gradient-boosting challenger, a cut-off strategy from the loan economics, fairness testing across protected groups and drift monitoring. Bad = 90+ days past due within 12 months of booking.

Overall assessment: Approved for use with conditions - 3 point(s) to address

11 tests passed, 3 warnings, 0 failed (table below). Findings are drafted from the figures and reviewed by the analyst before release.

GINI, OUT-OF-TIME	KS, OUT-OF-TIME	BAD RATE, DEVELOPMENT
0.441	**0.335**	**6.0%**
95% interval 0.41 to 0.47	AUC 0.721	1,297 bad loans

RECOMMENDED CUT-OFF	SCORE PSI, LATEST QUARTER	LOWEST FAIRNESS RATIO
555	**0.184**	**0.82**
approval 79.8%	2026Q3: WATCH	age_group = under 30

Key findings

1. The scorecard was developed on 21,772 booked loans with a mature outcome (1,297 bads, bad rate 6.0%) and validated on 12,742 later loans out-of-time (bad rate 6.0%). Bad = 90+ days past due within 12 months of booking.
2. Discrimination is good: Gini 0.441 out-of-time (95% interval 0.409 to 0.475), KS 0.335; training 0.465 and in-time test 0.460 show no material overfitting.
3. The gradient-boosting challenger reaches Gini 0.423 out-of-time (-0.018 versus the scorecard): the transparent scorecard gives up nothing material.
4. Calibration: observed bad rate is 0.99x the predicted PD out-of-time (Hosmer-Lemeshow p = 0.002; with samples this large small deviations are significant, so the ratio matters more).
5. The most influential characteristics (points range) are utilisation, credit_history_months, employment_years, inquiries_6m; 12 of 16 candidate characteristics were retained.
6. Recommended cut-off 555: approval rate 79.8%, bad rate of the approved book 4.0%; applicants within 15 points below it (12.0%) go to manual review (bad rate 11.1%).
7. Fairness: the lowest adverse impact ratio is 0.82 (age_group = under 30), status WARN. Re-fitting without credit_history_months lifts it to 0.93 at a cost of 0.020 Gini.
8. Drift: score PSI in 2026Q3 is 0.184 (WATCH); characteristics shifting: inquiries_6m (CSI 0.99), purpose (CSI 0.24).

Validation tests

Test	Result	Threshold	Status	Why it matters
Discrimination: Gini on out-of-time sample	0.441	>= 0.3 pass, >= 0.2 warn	PASS	Rank-ordering power on later applications that the model never saw.
Discrimination: KS on out-of-time sample	0.335	>= 0.25 pass, >= 0.15 warn	PASS	Largest separation between the cumulative distributions of goods and bads.
Stability: Gini drop from training to out-of-time	0.023	<= 0.05 pass, <= 0.1 warn	PASS	A large fall means the model fits the past better than the future (overfit or change in population).
Overfitting: Gini drop from training to in-time test	0.005	<= 0.07 pass, <= 0.12 warn	PASS	Same period, unseen applicants: measures fit to noise in the training sample.
Logic: all coefficients have the expected sign	yes	all negative on WoE	PASS	A wrong sign means a characteristic pushes risk the opposite way to the data: not defensible.
Multicollinearity: largest VIF	1.73	<= 5.0 pass, <= 10.0 warn	PASS	High VIF makes coefficients unstable.
Leakage screen: largest information value	0.169	<= 0.5 pass, <= 1.0 warn	PASS	A characteristic that is too good to be true often contains the outcome (information not available at application).
Sample: bad loans in the development sample	1,297	>= 500 pass, >= 300 warn	PASS	Few bads make coefficients and bins unreliable.
Calibration: observed vs predicted bad rate, out-of-time	0.99x	|ratio - 1| <= 0.2 pass, <= 0.35 warn	PASS	The scorecard's PD must match reality, or provisioning and pricing built on it will be wrong.
Drift: score PSI, latest period	0.184	<= 0.1 pass, <= 0.25 warn	WARN	Shift of the score distribution versus development.
Drift: largest characteristic CSI, latest period	0.988	<= 0.1 pass, <= 0.25 warn	WARN	Shift of the applicant mix on inquiries_6m. Breach: investigate and recalibrate or redevelop; monitoring action required.
Challenger: Gini advantage of the gradient-boosting model	-0.018	<= 0.03 pass, <= 0.06 warn	PASS	A large advantage means the scorecard misses interactions: investigate before relying on it.
Fairness: lowest adverse impact ratio across protected groups	0.82	>= 0.9 pass, >= 0.8 warn	WARN	Approval rate relative to the best-treated group (age_group = under 30).
Data: quality checks on the inputs	passed	no error-level issue	PASS	See the data-quality appendix.

Source: tables/validation_checks.csv

Conditions and recommended actions

- Investigate the shift in the score distribution; recalibrate or redevelop if it persists for two more quarters.
- Investigate the change in applicant mix on the characteristic named; check data feeds and marketing channels first.
- Review the flagged group with the fair-lending / compliance function; consider the less discriminatory alternative scorecard.

Methodology

This report tells how the scorecard performs and how it should be used. The steps, formulas, thresholds and references are in the methodology document. One thing helps when reading on: every headline result comes from the out-of-time sample, loans later than any the model was built on.

Results

1. Data and samples

Step	Applications
Applications received	68,252
Declined by the legacy policy	7,770
Booked loans	60,482
Outcome not yet mature (scored for monitoring only)	13,979
Mature but outcome missing (excluded)	140
Used to build and validate the model	46,363

Source: results.json

Sample	Loans	First application	Last application	Bads	Bad rate
excluded	4,539	2022-01-01	2024-12-31	0	n/a
monitoring	29,199	2025-01-01	2026-09-30	953	8.0%
oot	12,742	2024-01-01	2024-12-31	770	6.0%
test	6,531	2022-01-01	2023-12-31	389	6.0%
train	15,241	2022-01-01	2023-12-31	908	6.0%

Source: tables/samples.csv

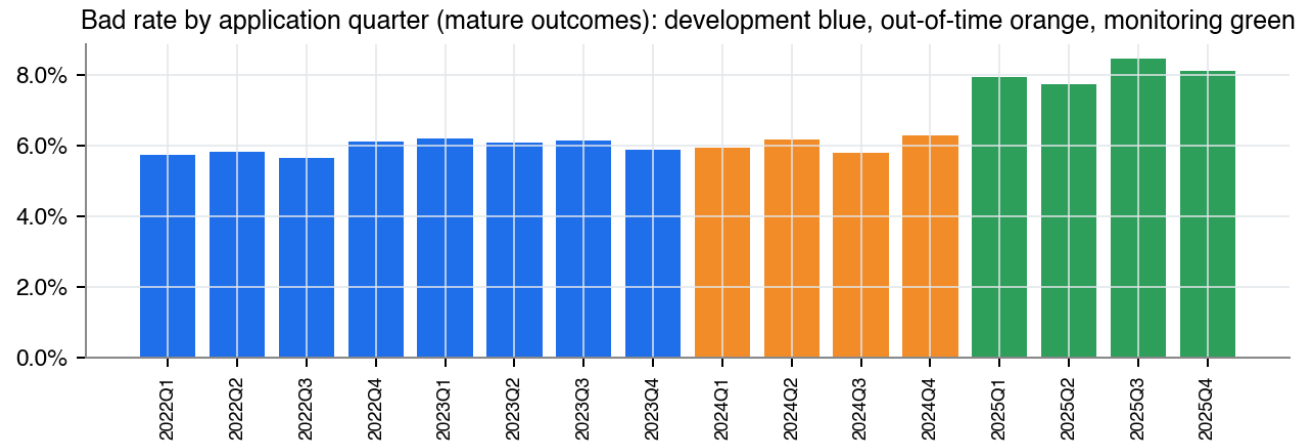


Figure 1. Bad rate by quarter. Source: interim/scored.parquet (mature outcomes)

2. Characteristics

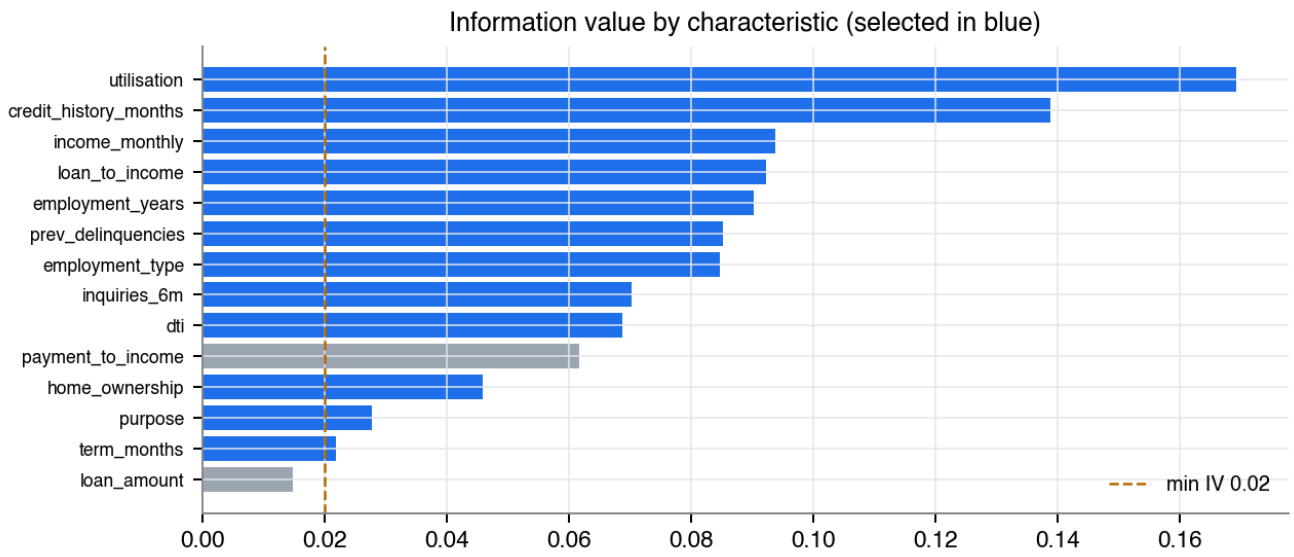


Figure 2. Information value. Source: tables/information_value.csv

Characteristic	Type	Bins	IV	Trend	Strength
utilisation	numeric	8	0.169	increasing	medium
credit_history_months	numeric	6	0.139	decreasing	medium
income_monthly	numeric	8	0.094	decreasing	weak
loan_to_income	numeric	7	0.092	increasing	weak
employment_years	numeric	7	0.090	decreasing	weak
prev_delinquencies	numeric	2	0.085	increasing	weak
employment_type	categorical	4	0.085	categorical	weak
inquiries_6m	numeric	4	0.070	increasing	weak
dti	numeric	6	0.069	increasing	weak
payment_to_income	numeric	6	0.062	increasing	weak
home_ownership	categorical	4	0.046	categorical	weak
purpose	categorical	6	0.028	categorical	weak
term_months	numeric	5	0.022	increasing	weak
loan_amount	numeric	4	0.015	increasing	useless

Source: tables/information_value.csv

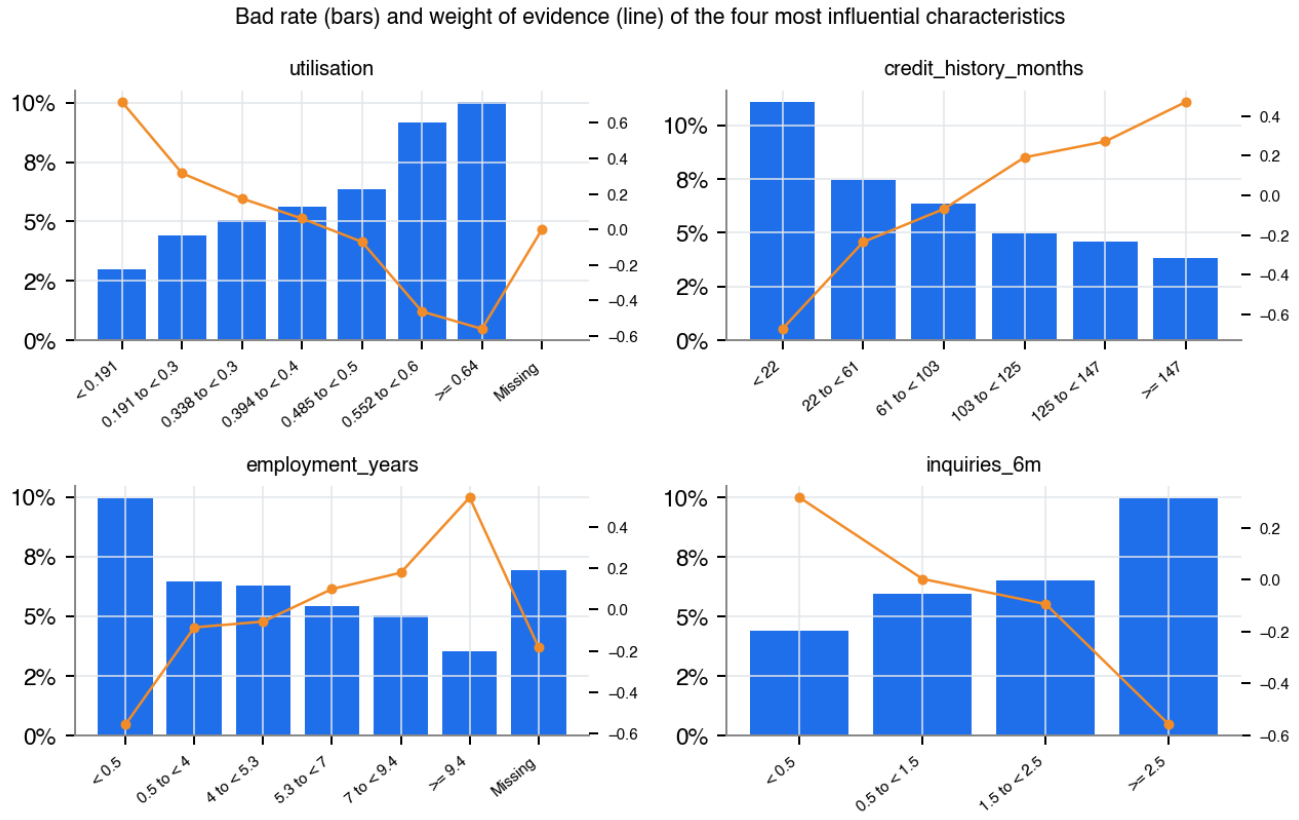


Figure 3. Bad rate and WoE by bin. Source: tables/woe_bins.csv

3. The scorecard

12 characteristics, 75 bins. A bin's points are added; the sum is the score (higher = safer). Characteristics are listed by influence (points range).

Characteristic	Coefficient	Std error	p-value	VIF	IV	Points range
utilisation	-1.013	0.086	0.000	1.003	0.169	37
credit_history_months	-0.869	0.094	0.000	1.026	0.139	29
income_monthly	-0.326	0.148	0.027	1.608	0.094	9
loan_to_income	-0.342	0.152	0.024	1.732	0.092	10
employment_years	-0.796	0.119	0.000	1.025	0.090	26
prev_delinquencies	-0.854	0.118	0.000	1.033	0.085	15
employment_type	-0.706	0.124	0.000	1.146	0.085	15
inquiries_6m	-0.994	0.129	0.000	1.001	0.070	25
dti	-0.803	0.155	0.000	1.344	0.069	14
home_ownership	-0.734	0.168	0.000	1.033	0.046	11
purpose	-1.063	0.207	0.000	1.001	0.028	13
term_months	-1.186	0.251	0.000	1.075	0.022	15

Source: tables/scorecard_stats.csv

Characteristic	Bin	Share	Bad rate	WoE	Points
utilisation	< 0.191	14.9%	3.0%	+0.72	68
	0.191 to < 0.338	24.9%	4.4%	+0.32	56
	0.338 to < 0.394	10.0%	5.0%	+0.17	52
	0.394 to < 0.485	15.0%	5.6%	+0.06	49
	0.485 to < 0.552	10.0%	6.4%	-0.07	45
	0.552 to < 0.64	10.0%	9.2%	-0.46	34
	>= 0.64	15.1%	10.0%	-0.56	31
	Missing	0.0%	0.0%	+0.00	47
credit_history_months	< 22	10.0%	11.1%	-0.67	30
	22 to < 61	19.9%	7.4%	-0.24	41
	61 to < 103	20.0%	6.4%	-0.07	45
	103 to < 125	9.9%	5.0%	+0.19	52
	125 to < 147	9.9%	4.6%	+0.27	54
	>= 147	30.3%	3.8%	+0.47	59
employment_years	< 0.5	9.2%	10.0%	-0.56	34
	0.5 to < 4	38.5%	6.5%	-0.09	45
	4 to < 5.3	10.3%	6.3%	-0.06	46
	5.3 to < 7	9.7%	5.4%	+0.10	49
	7 to < 9.4	10.5%	5.0%	+0.18	51
	>= 9.4	19.6%	3.5%	+0.54	60
	Missing	2.2%	6.9%	-0.18	43
inquiries_6m	< 0.5	33.0%	4.4%	+0.32	56
	0.5 to < 1.5	36.5%	6.0%	+0.00	47
	1.5 to < 2.5	20.6%	6.5%	-0.09	44
	>= 2.5	9.9%	10.0%	-0.56	31
prev_delinquencies	< 0.5	72.9%	4.9%	+0.21	52
	>= 0.5	27.1%	8.8%	-0.42	37
employment_type	Salaried	69.3%	4.9%	+0.20	51
	Self-employed	13.9%	6.7%	-0.12	45
	Part-time	15.1%	9.8%	-0.53	36
	Other (rare / unseen)	1.8%	8.1%	-0.35	40
term_months	< 18	7.6%	4.8%	+0.22	55
	18 to < 30	25.1%	5.0%	+0.19	54
	30 to < 42	32.6%	6.0%	-0.01	47
	42 to < 54	19.1%	6.4%	-0.08	44
	>= 54	15.6%	7.4%	-0.22	40

Characteristic	Bin	Share	Bad rate	WoE	Points
dti	< 0.2294	24.3%	4.6%	+0.27	53
	0.2294 to < 0.325	24.3%	4.7%	+0.25	53
	0.325 to < 0.386	14.6%	6.0%	-0.01	47
	0.386 to < 0.4341	9.7%	7.0%	-0.17	43
	>= 0.4341	24.3%	8.3%	-0.35	39
	Missing	2.9%	4.6%	+0.26	53
purpose	Medical	9.7%	5.0%	+0.18	53
	Other	13.4%	5.1%	+0.16	52
	Car	21.9%	5.3%	+0.12	51
	Education	8.0%	5.6%	+0.06	49
	Home improvement	19.9%	5.7%	+0.05	49
	Debt consolidation	27.1%	7.5%	-0.24	40
home_ownership	Own	22.1%	4.4%	+0.31	54
	Mortgage	33.7%	5.3%	+0.13	50
	Rent	39.2%	7.3%	-0.22	43
	Other (rare / unseen)	5.1%	6.6%	-0.12	45
loan_to_income	< 0.1502	24.3%	3.9%	+0.45	52
	0.1502 to < 0.2016	14.6%	5.3%	+0.12	48
	0.2016 to < 0.2402	9.7%	5.8%	+0.02	47
	0.2402 to < 0.3388	19.4%	6.1%	-0.03	47
	0.3388 to < 0.4656	14.6%	6.6%	-0.11	46
	>= 0.4656	14.6%	9.5%	-0.50	42
	Missing	2.9%	4.6%	+0.26	50
income_monthly	< 2,059	19.4%	8.8%	-0.42	43
	2,059 to < 2,390	9.7%	7.2%	-0.21	45
	2,390 to < 2,881	14.6%	6.5%	-0.10	46
	2,881 to < 3,216	9.7%	5.6%	+0.06	48
	3,216 to < 3,855	14.6%	5.0%	+0.19	49
	3,855 to < 5,390	19.4%	4.3%	+0.35	50
	>= 5,390	9.7%	3.7%	+0.49	52
	Missing	2.9%	4.6%	+0.26	50

Source: tables/scorecard.csv

Selection log

- loan_amount: dropped, IV 0.015 < 0.02
- region: dropped, IV 0.011 < 0.02
- existing_monthly_debt: dropped, IV 0.010 < 0.02

• payment_to_income: dropped, correlation 0.82 with loan_to_income (higher IV kept)

4. Discrimination

Model	Sample	Loans	Bads	Bad rate	AUC	Gini	Gini 95% interval	KS
Scorecard	train	15,241	908	6.0%	0.732	0.465	0.428 to 0.498	0.354
Challenger (gradient boosting)	train	15,241	908	6.0%	0.802	0.603	0.571 to 0.631	0.455
Scorecard	test	6,531	389	6.0%	0.730	0.460	0.412 to 0.508	0.355
Challenger (gradient boosting)	test	6,531	389	6.0%	0.732	0.464	0.407 to 0.510	0.355
Scorecard	oot	12,742	770	6.0%	0.721	0.441	0.409 to 0.475	0.335
Challenger (gradient boosting)	oot	12,742	770	6.0%	0.711	0.423	0.387 to 0.458	0.305

Source: tables/performance_summary.csv

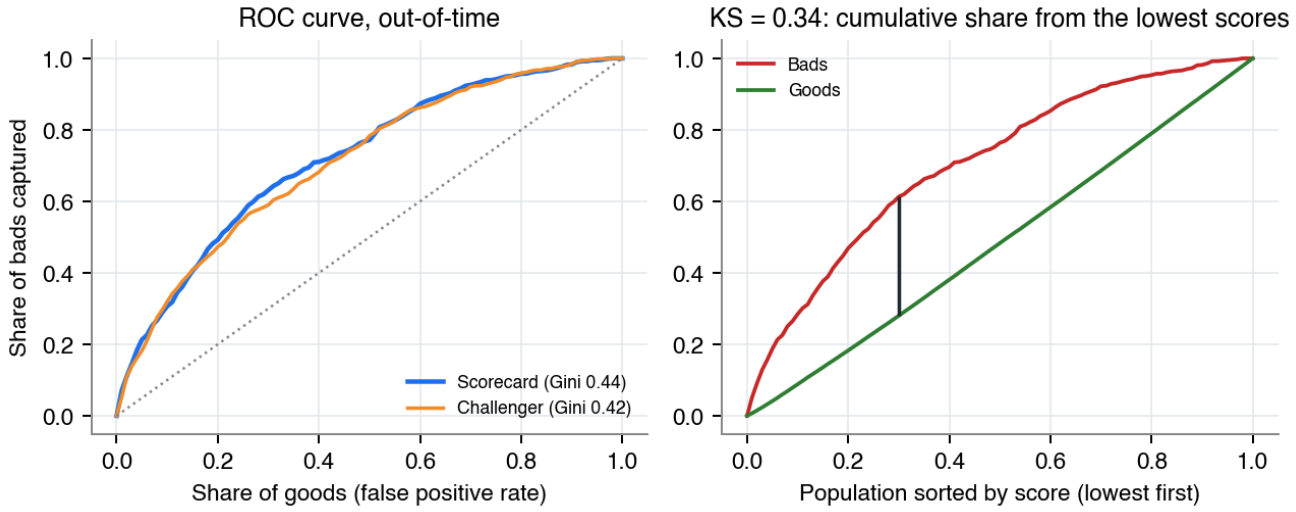


Figure 4. ROC and KS, out-of-time. Source: tables/roc_oot.csv, tables/ks_curve_oot.csv

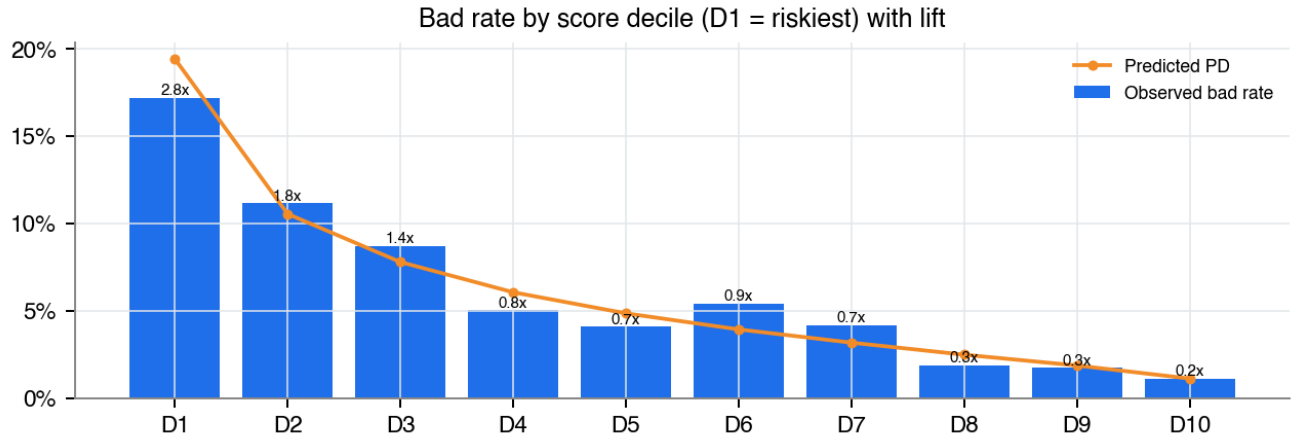


Figure 5. Lift by decile. Source: tables/lift_deciles.csv

Decile	Loans	Bads	Bad rate	Predicted PD	Cumulative bads captured	Lift
1	1275	219	17.2%	19.4%	28.4%	2.84
2	1274	142	11.1%	10.5%	46.9%	1.84
3	1274	111	8.7%	7.8%	61.3%	1.44
4	1274	64	5.0%	6.0%	69.6%	0.83
5	1274	52	4.1%	4.8%	76.4%	0.68
6	1274	69	5.4%	3.9%	85.3%	0.90
7	1274	53	4.2%	3.2%	92.2%	0.69
8	1274	24	1.9%	2.5%	95.3%	0.31
9	1274	22	1.7%	1.9%	98.2%	0.29
10	1275	14	1.1%	1.1%	100.0%	0.18

Source: tables/lift_deciles.csv

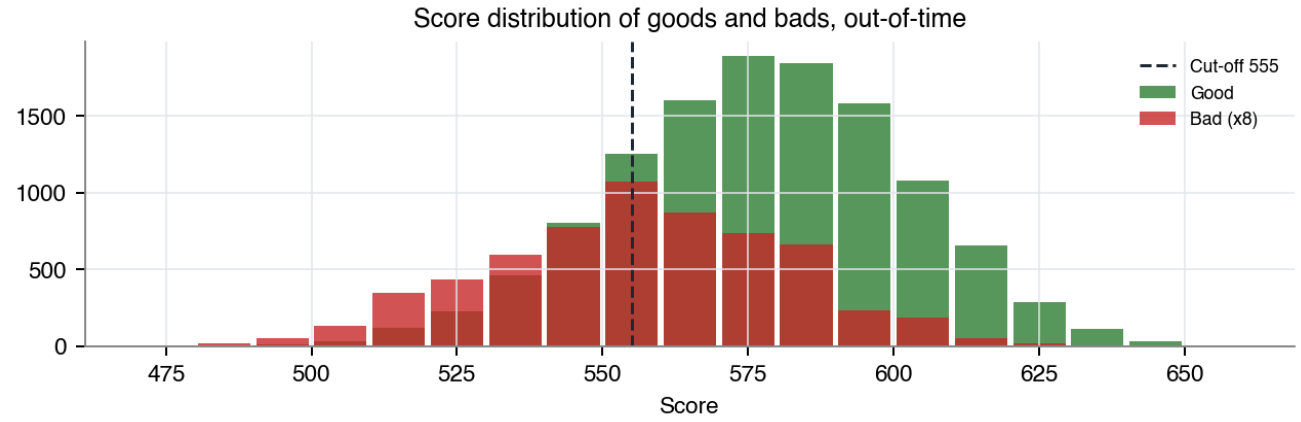


Figure 6. Score distribution. Source: tables/score_distribution_oot.csv

5. Calibration, PD and risk grades

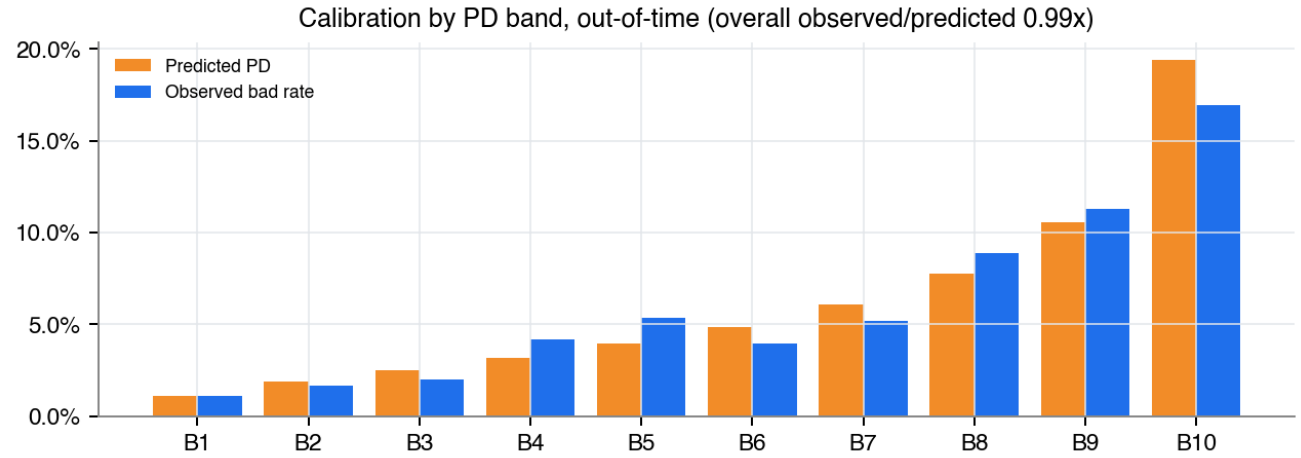


Figure 7. Predicted vs observed bad rate. Source: tables/calibration_oot.csv

Band	Loans	Bads	Predicted PD	Observed	Obs/Pred	p-value
1	1275	14	1.11%	1.10%	0.99	1.10
2	1274	21	1.85%	1.65%	0.89	0.68
3	1274	25	2.47%	1.96%	0.79	0.27
4	1274	53	3.16%	4.16%	1.31	0.06
5	1274	68	3.92%	5.34%	1.36	0.02
6	1274	50	4.84%	3.92%	0.81	0.14
7	1274	66	6.05%	5.18%	0.86	0.21
8	1274	113	7.78%	8.87%	1.14	0.17
9	1274	144	10.54%	11.30%	1.07	0.40
10	1275	216	19.41%	16.94%	0.87	0.03

Source: tables/calibration_oot.csv

Risk grades (out-of-time)

Grade	Loans	Bads	Bad rate	Predicted PD	Min score	Max score	Share
A	2647	38	1.4%	1.5%	597	656	20.8%
B	2482	76	3.1%	2.9%	583	596	19.5%
C	2578	121	4.7%	4.4%	570	582	20.2%
D	2596	188	7.2%	7.0%	554	569	20.4%
E	2439	347	14.2%	15.2%	472	553	19.1%

Source: tables/risk_grades.csv

Score to PD

Score	PD	Good:bad odds
400	95.34%	0 : 1
440	83.66%	0 : 1
480	56.14%	1 : 1
520	24.24%	3 : 1
560	7.41%	12 : 1
600	1.96%	50 : 1
640	0.50%	200 : 1
680	0.12%	800 : 1

Source: tables/score_to_pd.csv

6. Challenger and explainability

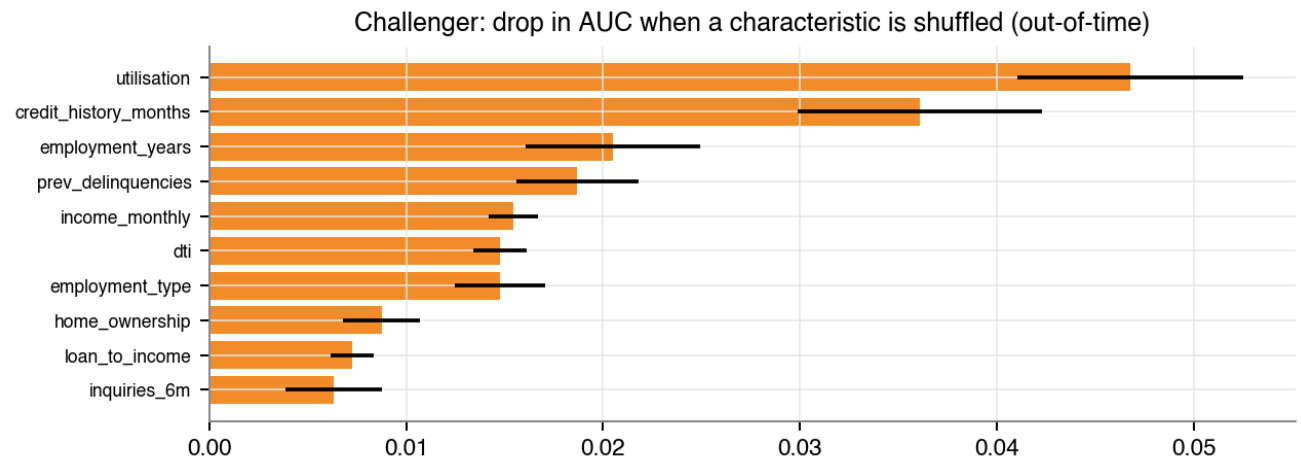


Figure 8. Challenger importance. Source: tables/challenger_importance.csv

The scorecard explains every decision characteristic by characteristic. For a declined applicant the reasons are the characteristics that cost the most points compared with the best bin (adverse-action reasons). Among the 2,572 out-of-time applicants below the cut-off:

Characteristic	Mentions	Share of declined	Principal reason
utilisation	2,245	87.3%	1,674
employment_years	1,929	75.0%	183
credit_history_months	1,824	70.9%	498
prev_delinquencies	1,248	48.5%	19
employment_type	784	30.5%	7
inquiries_6m	667	25.9%	191
dti	618	24.0%	0
term_months	443	17.2%	0

Source: tables/adverse_reasons.csv

7. Cut-off strategy

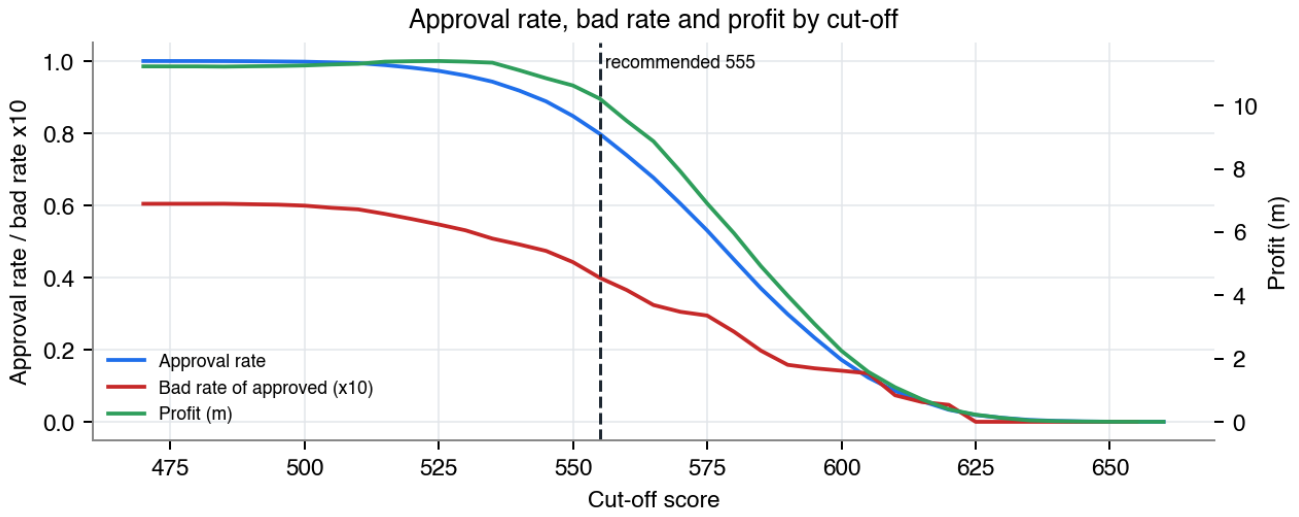


Figure 9. Approval, bad rate and profit by cut-off. Source: tables/cutoff_table.csv

Recommended cut-off **555**: approval rate 79.8%, bad rate of the approved book 3.99% (the bad-rate appetite is binding; the unconstrained profit optimum would be 525). Applicants within 15 points below it (12.0%, bad rate 11.1%) are referred to manual review.

Cut-off	Approved	Approval rate	Bad rate (approved)	Bads approved	Profit	Profit per applicant	Bad rate (declined)
525	12,395	97.3%	5.47%	678	11.41m	895	26.5%
530	12,230	96.0%	5.31%	649	11.39m	894	23.6%
535	12,013	94.3%	5.08%	610	11.36m	891	21.9%
540	11,697	91.8%	4.92%	575	11.12m	873	18.7%
545	11,316	88.8%	4.74%	536	10.86m	853	16.4%
550	10,800	84.8%	4.43%	478	10.63m	835	15.0%
555	10,170	79.8%	3.99%	406	10.22m	802	14.2%
560	9,417	73.9%	3.65%	344	9.52m	747	12.8%
565	8,620	67.7%	3.24%	279	8.87m	696	11.9%
570	7,707	60.5%	3.05%	235	7.92m	621	10.6%
575	6,757	53.0%	2.95%	199	6.91m	542	9.5%
580	5,728	45.0%	2.50%	143	5.96m	468	8.9%
585	4,715	37.0%	1.97%	93	4.92m	386	8.4%

Source: tables/cutoff_table.csv

Sensitivity to the economics

LGD	Net margin	Cut-off (with appetite)	Cut-off (no appetite limit)	Approval rate	Profit
55.0%	6.0%	555	535	79.8%	6.36m
55.0%	9.0%	555	515	79.8%	10.61m
55.0%	12.0%	555	515	79.8%	14.86m
65.0%	6.0%	555	535	79.8%	5.97m
65.0%	9.0%	555	525	79.8%	10.22m
65.0%	12.0%	555	515	79.8%	14.47m
75.0%	6.0%	555	550	79.8%	5.57m
75.0%	9.0%	555	535	79.8%	9.83m
75.0%	12.0%	555	515	79.8%	14.08m

Source: tables/cutoff_sensitivity.csv

8. Fairness

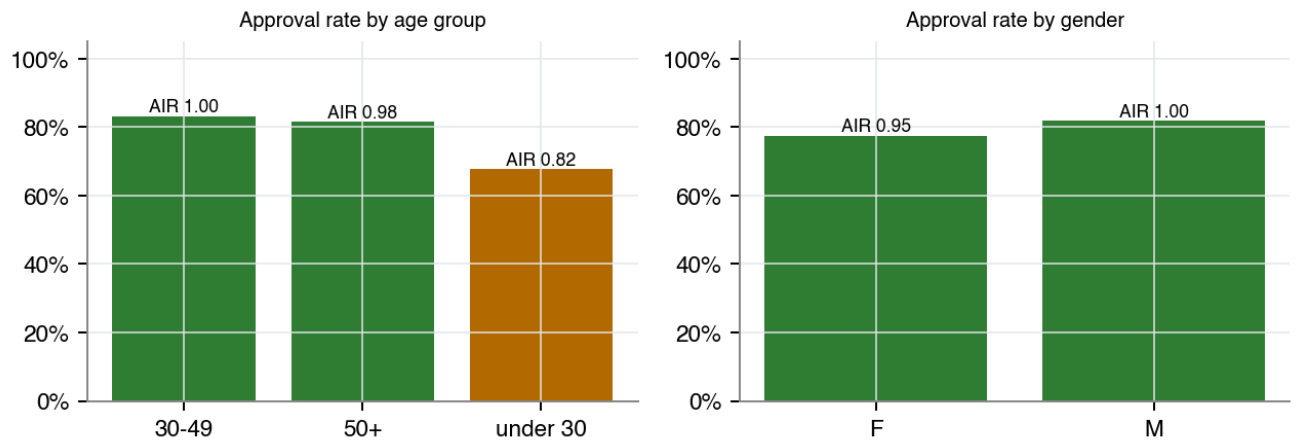


Figure 10. Approval rate and adverse impact ratio by group at the cut-off. Source: tables/fairness_groups.csv

Attribute	Group	Loans	Approval	AIR	Bad rate	Obs/Pred	AUC	Goods approved	Bads approved	Status
gender	F	6,019	77.5%	0.95	6.5%	1.00	0.725	79.4%	50.1%	PASS
gender	M	6,723	81.9%	1.00	5.6%	0.98	0.714	83.4%	55.4%	PASS
age_group	30-49	8,208	83.2%	1.00	4.8%	0.87	0.700	84.3%	62.6%	PASS
age_group	50+	1,890	81.6%	0.98	6.6%	1.14	0.731	83.7%	51.6%	PASS
age_group	under 30	2,644	67.9%	0.82	9.5%	1.15	0.719	71.0%	37.6%	WARN

Source: tables/fairness_groups.csv

AIR = approval rate relative to the best-treated group; below 0.80 is the "four-fifths" flag. Disparity is not proof of discrimination and a pass is not a legal opinion; standards depend on the jurisdiction and product.

Proxy analysis

Attribute	Characteristic	Direct use	Cramer's V	Association
age_group	credit_history_months	False	0.27	moderate
age_group	employment_type	False	0.22	moderate
gender	employment_type	False	0.22	moderate
age_group	employment_years	False	0.16	weak
gender	income_monthly	False	0.09	negligible
gender	loan_to_income	False	0.05	negligible
age_group	prev_delinquencies	False	0.04	negligible
age_group	income_monthly	False	0.03	negligible

Source: tables/fairness_proxies.csv

Less discriminatory alternative

Scorecard	Dropped	Characteristics	Cut-off	Approval rate	Gini	KS	Group	AIR (worst group)	Lowest AIR
Current scorecard		12	555	79.8%	0.441	0.335	under 30	0.82	0.82
Alternative without credit_history_months	credit_history_months	11	556	80.2%	0.421	0.314	under 30	0.93	0.93

Source: tables/fairness_alternative.csv

9. Monitoring

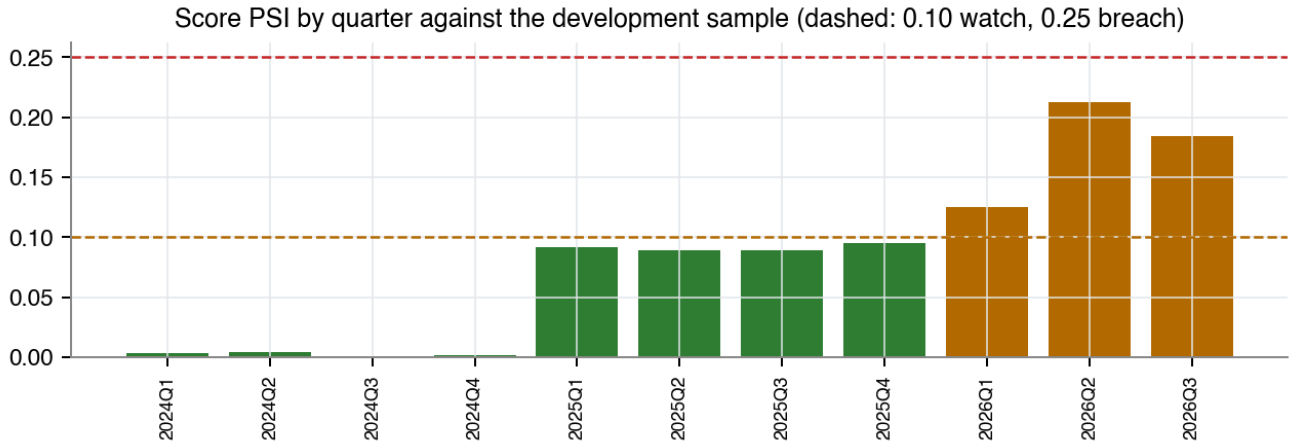


Figure 11. Score PSI by quarter. Source: tables/psi_score.csv

Quarter	Applications	Mean score	PSI	Status
2024Q1	3,048	575.6	0.004	PASS
2024Q2	3,263	575.6	0.004	PASS
2024Q3	3,147	575.8	0.001	PASS
2024Q4	3,284	574.8	0.002	PASS
2025Q1	3,836	567.6	0.092	PASS
2025Q2	3,954	568.0	0.089	PASS
2025Q3	4,200	567.8	0.089	PASS
2025Q4	4,160	567.5	0.095	PASS
2026Q1	4,342	566.2	0.125	WARN
2026Q2	4,276	563.4	0.212	WARN
2026Q3	4,431	564.2	0.184	WARN

Source: tables/psi_score.csv

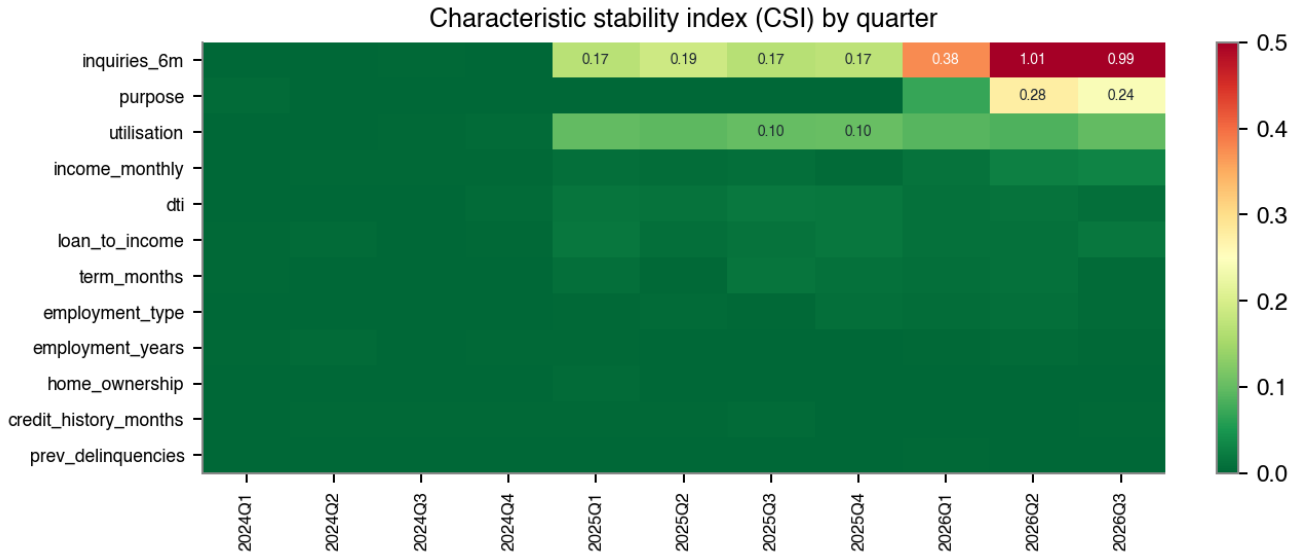


Figure 12. CSI by characteristic and quarter. Source: tables/csi.csv

Categories not seen in development

Characteristic	Quarter	Unseen share	Values
purpose	2026Q1	1.4%	Crypto
purpose	2026Q2	4.5%	Crypto
purpose	2026Q3	4.0%	Crypto

Source: tables/new_categories.csv

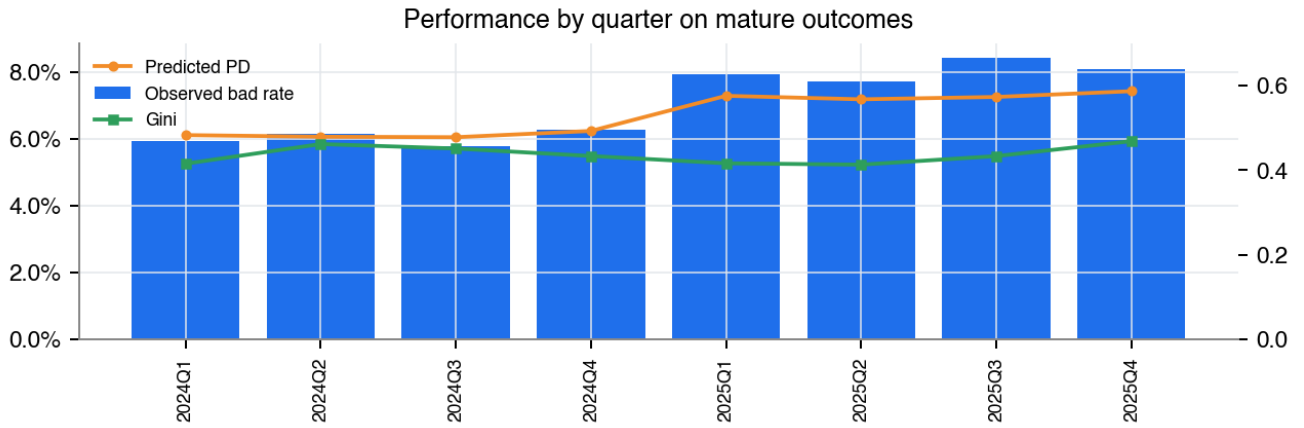


Figure 13. Performance by quarter (mature outcomes). Source: tables/performance_by_period.csv

Quarter	Loans	Bads	Bad rate	Predicted PD	Obs/Pred	Gini
2024Q1	3,048	181	5.9%	6.1%	0.97	0.415
2024Q2	3,263	201	6.2%	6.1%	1.02	0.462
2024Q3	3,147	182	5.8%	6.0%	0.96	0.451
2024Q4	3,284	206	6.3%	6.2%	1.01	0.433
2025Q1	3,364	267	7.9%	7.3%	1.09	0.416
2025Q2	3,514	271	7.7%	7.2%	1.07	0.413
2025Q3	3,698	312	8.4%	7.3%	1.16	0.433
2025Q4	1,273	103	8.1%	7.4%	1.09	0.469

Source: tables/performance_by_period.csv

Limitations and assumptions

- Outcomes exist only for loans the lender granted. No reject inference is applied, so the model describes the through-the-door population only within the historically accepted region; cut-offs below it are extrapolations.
- The bad definition (90+ days past due within 12 months of booking) and the outcome window are fixed; a different definition requires a new development.
- The cut-off economics (margin, LGD) are the lender's assumptions; the recommended cut-off is only as good as they are (see sensitivity).
- Fairness is tested on the protected attributes available in the data; other groups and intersections are not covered. The analysis is statistical, not a legal opinion.
- Drift measures change in the applicant mix, not model failure: performance monitoring on mature outcomes lags by 12 months.
- Hosmer-Lemeshow and other significance tests reject small deviations in large samples; the ratio of observed to predicted bad rate matters more.
- This demonstration uses synthetic data; its numbers have no meaning for any real lender.

Appendix

A.1 Configuration used

Samples	development 2022-01-01 to 2023-12-31; out-of-time 2024-01-01 to 2024-12-31; monitoring from 2025-01-01
Binning	max bins 7, min bin share 0.06
Selection	IV ≥ 0.02, |corr| ≤ 0.7, p ≤ 0.05
Scaling	score 600 at 50:1 odds, 20 points to double the odds (factor 28.85, offset 487.12)
Economics	net margin 9.0%, LGD 65%, bad-rate appetite 4.0%
Monitoring thresholds	PSI / CSI watch 0.1, breach 0.25

Source: config_used.yaml in the run folder

A.2 Data cleaning log

step	action	rows_affected	key
applications	drop exact duplicate application rows	1	
applications	category spelling harmonised (case / spaces)	60	home_ownership
applications	impossible values set to missing (outside [18, 100])	2	age
applications	impossible values set to missing (outside [1, None])	2	income_monthly
applications	impossible values set to missing (outside [0, 1.5])	3	utilisation
performance	outcome rows without a matching application (dropped)	3	
performance	booked loans past the outcome window without an outcome (excluded; ask the lender)	140	
samples	applications with a mature outcome used for model building and validation	46363	
samples	booked loans with an immature outcome (not used; scored for monitoring only)	13979	

Source: tables/cleaning_log.csv

Data quality

applications · 68253 rows · PASSED (0 errors, 5 warnings)

Check	Severity	Column	Count	Message
missing_values	info	employment_years	1427	1427 missing (2.09%)
missing_values	info	income_monthly	1908	1908 missing (2.80%)
duplicate_rows	warning		1	1 fully duplicated rows
duplicate_keys	warning		2	2 rows share a key ['application_id'] (0 with conflicting values)
out_of_range	warning	age	2	2 values outside the business range [18, 100]
out_of_range	warning	income_monthly	2	2 values outside the business range [1, None]
out_of_range	warning	utilisation	3	3 values outside the business range [0, 1.5]

Source: data_quality/dq_applications.json

applications_clean · 68252 rows · PASSED (0 errors, 2 warnings)

Check	Severity	Column	Count	Message
missing_values	warning	age	2	2 missing (0.00%)
missing_values	info	employment_years	1427	1427 missing (2.09%)
missing_values	info	income_monthly	1909	1909 missing (2.80%)
missing_values	warning	utilisation	3	3 missing (0.00%)
missing_values	info	default_12m	21889	21889 missing (32.07%)

Source: data_quality/dq_applications_clean.json

performance · 46366 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

Audit trail

Run ID	`cs_20261111T175902Z_9423c6`
Run started / finished (UTC)	2026-11-11T17:59:02+00:00
Analyst	info@proquantanalytics.com
Code version (git)	`2703b230a84f (uncommitted changes) · core 0.1.0`
Config SHA-256	`3c204b428b9c48cb`
Input: applications	`68253 rows · sha256 4066bc75c1d0ce2c`
Input: performance	`46366 rows · sha256 b728ddc6477b9f52`
Environment	`python 3.13.5 · pandas 2.2.3 · numpy 2.1.3`

Source: audit.json in the run folder

This report was prepared by ProQuant Analytics for Demo Client under the terms of the engagement agreement, from data provided by the lender, using the methodology described herein. Statistical models describe past behaviour and can be wrong in new conditions. The report does not constitute legal, regulatory, credit or investment advice.